

The wage-gap brief

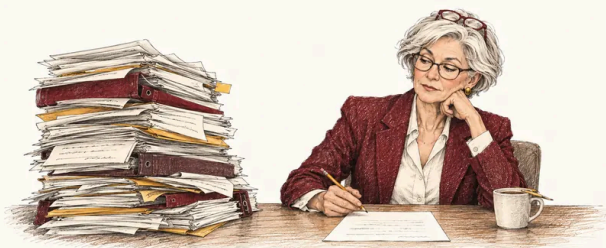
This outline lists all the pages in the problem set, to give you an overview. Text in *italics* denotes fields that the quiz will fill in from your dataset.

THE REQUEST

The job

The chair of a U.S. federal taskforce on women's and men's earnings has hired your consultancy. The taskforce advises Congress, and the chair wants your answer to two questions: how large is the difference between what women and men who work full time earn per hour, and what can the taskforce conclude from that difference?

The chair will quote your findings to Congress, so you must be careful about what you recommend and make sure that you don't claim results beyond what you can conclude from your regressions.



The job has four parts:

Part of the job	What you'll do
The analysis	You'll analyze a sample of US workers in a Colab notebook and compute some summary statistics and run a few regressions with hourly wages. You'll enter your results in question 1 and interpret your findings.
Your recommendation	You'll be asked to decide which of your estimates the chair should report, and recommend how the chair should describe your findings. This part will need you to think carefully about what your numbers can tell us. After that, the chair might ask you a follow-up question.
The meeting	You'll present your findings to the taskforce. The meeting will have five people who will ask you questions: an economic adviser, a taskforce member, a journalist, a colleague from your firm (who's run the same analysis on a different sample), and an analyst at the taskforce. After answering their questions, you'll get a chance to revise your advice to the chair.
The memo (question 2)	You'll write a short memo to the chair, based on the results and reasons listed on the last page of question 1. You must attach your Colab notebook to the memo.

An analyst at the taskforce has your dataset and can check your calculations for you. If you are unable to obtain the correct regression results, you'll be able to ask to see the analyst's estimates instead. At the meeting, the analyst will also show results from a separate sample of 500 workers.

Once you send your recommendation to the chair, it is locked, along with your sample and your chosen estimate. Once you send your revised recommendation after the meeting, it is locked too. Before each of these two sends, you'll tick a box and then confirm the send.

We recommend working through question 1 over several sittings: work through the Colab analysis before returning to the quiz to enter your results. Everything you enter saves as you go, and the parts of question 1 you finish earn their marks even if you don't finish. You can close this tab and come back later.

Some of the questions on the final exam will draw on the reasoning in this problem set, based on results for a different random sample of workers. If you think carefully about your choices in the problem set, you'll be in good shape to answer those questions.

[Your dataset →](#)

THE REQUEST

Your dataset

Your dataset comes from the Current Population Survey (CPS), a monthly survey of US households run by the US Census Bureau and the Bureau of Labor Statistics. Read more about the CPS: <https://www.census.gov/programs-surveys/cps.html>

One row in your dataset is one person. Your dataset contains a random sample of 2,000 people aged 18 to 64 (inclusive) who work full time. We count a person as working full time if they usually work 30 or more hours per week at their main job.

We've given every student a different sample, so you can discuss your work with your peers, but don't expect to get the same numbers!

For each person, your dataset has these variables:

- **wage** : the person's wage in US dollars per hour, computed as their usual weekly earnings divided by their usual weekly hours (the hours that they report usually working each week at their main job)
- **educ** : the person's years of schooling
- **female** : takes on a value of 1 if the person is a woman, and 0 otherwise
- **age** : the person's age in years
- **occ** : the person's broad occupation group:
 - management and professional (1)
 - service (2)
 - sales and office (3)
 - construction and maintenance (4)
 - production and transport (5)

- `married` : takes on a value of 1 if the person is married and lives with their spouse, and 0 otherwise
- `kids` : takes on a value of 1 if the person's own children live in the household, and 0 otherwise
- `gov` : takes on a value of 1 if the person works for a federal, state, or local government, and 0 otherwise
- `union` : takes on a value of 1 if the person is a member of a labor union, and 0 otherwise
- `state` : the person's state of residence, as a FIPS code (question 1 lists the codes)

[Download my dataset \(CSV\)](#)

[The analysis →](#)

THE ANALYSIS

The analysis

The chair has sent your consultancy this request:



The chair

"Before I present anything to Congress, I'd like to know two things: First, who are we describing? I want to know things like how many workers your dataset contains, what women and men earn per hour on average, how many years of schooling each gender has, and what share of the workers in management and professional jobs are women. Second, I want to understand the size of the gap. Here, I think it might make sense to look at three comparisons: compare all women with all men, compare women and men who have the same years of schooling, and compare women and men with the same years of schooling who are also in the same broad occupation group. Tell me in plain words what each comparison shows. I'll probably have a follow-up question once I've read your results."

You'll do the analysis in a Colab notebook that we've set up for this problem set, the same way you work through the tutorials:

- Make sure that you're logged in to Google Colab
- Open the problem set notebook (<https://emiliatjernstrom.com/econ2041/problem-set/notebook>), making sure to work in a copy (click "Copy notebook" when prompted) as you need to save the notebook and upload it when you're done
- Upload your dataset to the notebook, following the instructions under "Load your data"

Keep your notebook. You must attach it to your memo in question 2, where it's worth 5 marks, so when you finish, download it from Colab (File → Download → Download .ipynb) and save it somewhere safe.

Use Python in your notebook for every calculation, and estimate every regression by OLS with `ols` from `statsmodels`, as in the tutorials. Apart from your choice of age range to include, use every row of your dataset.

On the next page, you'll choose which workers you want to include in your analysis. On the pages that follow, you'll see the quantities that you should calculate and estimate for the chair.

Your sample →

THE ANALYSIS

Your sample

On this page, you'll choose which group of US workers you aim to describe: full-time workers aged 18 to 64, or only the prime-age full-time workers aged 25 to 54. Your sample is the workers in your dataset who fall in that age range, and you'll use it to estimate the wage gap in the whole group, so you're deciding whose wage gap you'll report to the taskforce.

Both choices are defensible. A good reason points to something specific in your dataset or in the chair's request, in your own words. You can come back and change your choice until you send your recommendation to the chair.

Before you choose, use Python to calculate how many people in your dataset are aged 18 to 64 and how many are aged 25 to 54 (both ranges inclusive).

Which people will your analysis describe? (1 mark)

Full-time workers (aged 18 to 64 inclusive)

Prime-age full-time workers (aged 25 to 54 inclusive)

Why does this age range suit the question the taskforce is asking about the size of the wage gap?

0 / 600 characters

Describing your sample →

THE ANALYSIS

Describing your sample

Use Python in your notebook to calculate each number below for the workers in your sample, and enter it rounded to the decimals shown (the notebook reminds you how to round). Press Check after each one: you won't be able to move to the next page until you've checked all your numbers (though you can move through even if you haven't gotten the numbers right yet). For incorrect answers, you'll see a list of common issues, and you can keep trying as many times as you want.

Pressing Check will tell you whether what you've entered is correct, based on your dataset.

How many workers are in your sample (*full-time workers aged 18 to 64 or 25 to 54 (the ages you chose)*)? (1 mark)

N (0 decimals)

Check

What is the mean hourly wage of women and of men in your sample? (2 marks)

Women, US dollars per hour (2 decimals)

Men, US dollars per hour (2 decimals)

Check

What is the mean years of schooling of women and of men in your sample? (2 marks)

Women, years (1 decimal)

Men, years (1 decimal)

Check

What share of the workers in management and professional (`occ = 1`) in your sample are women? Enter a proportion between 0 and 1, not a percentage. (1 mark)

Proportion who are women (2 decimals)

Check

Estimating the gap →

THE ANALYSIS

Estimating the gap

Use Python in your notebook to estimate every regression on this page by OLS with `ols` from `statsmodels`, using every worker in your sample, and enter the results rounded to the decimals shown. Press Check after each one: you won't be able to move to the next page until you've checked all your numbers and written the three interpretations (though you can move through even if you haven't gotten the numbers right yet). For incorrect answers, you'll see a list of common issues, and you can keep trying as many times as you want.

Pressing Check will tell you whether what you've entered is correct, based on your dataset.

The three coefficients on `female` matter for everything that comes after, as does the p-value of the follow-up regression on a later page. You can leave this page with a wrong number, but to

continue past the Which estimate? page, you'll need each coefficient on **female** either entered correctly or replaced by the analyst's estimate. In the same way, you'll need the p-value entered correctly, or the analyst's p-value, before you tell the chair what it means.

If you can't get one of these numbers right, you'll be able to ask to see the analyst's number for your dataset. Each number has its own button, and asking for one doesn't show you the others. If you ask for the analyst's number, you can get at most half the marks for that number. If you are getting close to the deadline and you still aren't getting the right numbers, you may be better off asking to see the analyst's numbers so that you don't get a zero on those questions.

The later pages, including the meeting, show the analyst's estimates for your dataset, which are the same as yours once you've entered them correctly.

- ☐ I understand that if I ask to see the analyst's number for a coefficient or a p-value, I can get at most half the marks for that number.

Estimate a simple regression of **wage** on **female**, for *full-time workers aged 18 to 64 or 25 to 54 (the ages you chose), 30 or more hours a week*. What is the coefficient on **female**? What is the R squared of the regression? (2 marks)

Coefficient on **female**, dollars per hour (2 decimals)

R squared (3 decimals)

Check

Interpret the coefficient on **female**. State who the estimate describes, the direction and size of the difference, and the unit.

Your interpretation (1 mark)

0 / 600 characters

Estimate a regression of `wage` on `female` and years of schooling (`educ`), for the same workers. What is the coefficient on `female`? (1 mark)

Coefficient on `female`, dollars per hour (2 decimals)

Check

Interpret the coefficient on `female`. State who the estimate describes, the direction and size of the difference, and the unit.

Your interpretation (1 mark)

0 / 600 characters

Estimate a regression of `wage` on `female`, years of schooling (`educ`), and dummies for the broad occupation groups, with management and professional (`occ` = 1) as the reference category, for the same workers. What is the coefficient on `female`? (1 mark)

Coefficient on `female`, dollars per hour (2 decimals)

Check

Interpret the coefficient on `female`. State who the estimate describes, the direction and size of the difference, and the unit.

Your interpretation (1 mark)

0 / 600 characters

Choose the estimate →

THE ANALYSIS

Which estimate?

The table below lists the coefficient on `female` from each of your three regressions. As you know, adding controls changes our interpretation of the estimated coefficients, but adding more controls does not necessarily produce a "better" or "truer" estimate of the wage gap.

Specification	Coefficient on female (dollars per hour)	The comparison
No controls	<i>your coefficient</i>	All women with all men in your sample
Years of schooling	<i>your coefficient</i>	Women and men with the same years of schooling
Years of schooling and broad occupation-group dummies	<i>your coefficient</i>	Women and men with the same years of schooling in the same broad occupation group

A COEFFICIENT THAT DOES NOT MATCH YOUR DATASET YET SHOWS

Your coefficient isn't correct yet. Go back to the previous page and try again, or ask below to see the analyst's estimate.

- ☐ I understand that if I see the analyst's number, I can get at most half the marks for this number, even if I later enter it correctly myself.

Show the analyst's estimate

Which comparison should the chair report? (2 marks)

Compare all women with all men in your sample

Compare women and men with the same years of schooling

Compare women and men with the same years of schooling in the same broad occupation group

First, why is this the most useful comparison for the taskforce? Second, what is one limitation of this comparison?

0 / 600 characters

Your recommendation →

YOUR RECOMMENDATION

Your recommendation to the chair



The chair

"What did you find? Tell me which gap I should report to Congress and exactly what it compares."

Your estimated regression coefficient on **female** is *your coefficient* dollars per hour, *with the controls in the estimate you chose, if any.*

The finding that you recommend the chair report (1 mark)

The options appear in question 1.

- ☐ I understand that once I send this recommendation to the chair, the recommendation, my sample, and my chosen estimate are locked, even if I go back to an earlier page. (1 mark)

Send to the chair →

YOUR RECOMMENDATION

A follow-up question from the chair

The taskforce has agreed to use p-values below 0.1 as evidence against equal gaps.



The chair

"Thank you for your recommendation. One more thing before the meeting: several taskforce members have asked me whether the wage gap is the same for everyone. Pick one of their questions to answer, and make sure to tell me how confident you are that any differences are real and not just due to noise in the data."

Which question will you answer?

Does the gap differ by age?

Does the gap differ between workers who live with their spouse and those who don't?

Does the gap differ between workers with and without their own children living in the household?

Regress **wage** on the relevant terms, for your chosen sample of full-time workers. You should not include interactions other than the one that you selected above. You can add other variables as controls if you want to.

Tick every term in your chosen regression and type the estimated coefficient. The regression package will add an intercept automatically, so there's no box for it below.

(3 marks)

Column	In the regression	Its interaction with female
female	☐	
educ	☐	☐
age	☐	☐
occ	☐	☐
married	☐	☐
kids	☐	☐

gov

☐☐

union

☐☐

(3 marks)

EACH TERM YOU TICK GETS A BOX FOR ITS COEFFICIENT

Coefficient on *the term (its units, 2 decimals)*

Check

Interpret the three coefficients that are most relevant to the question. State who the estimates describe, the direction and size of each, along with the appropriate units.

Your interpretation (1 mark)

0 / 1500 characters

Which term in your regression do you need to answer the chair's question? (3 marks)

The terms you ticked appear here as options.

What is the p-value on that term in your regression table? (1 mark)

p-value (3 decimals)

Check

What should you tell the chair about that p-value? (1 mark)

The options appear in question 1.

The meeting →

The meeting



Economic adviser: economist who advises the taskforce on how to read statistical evidence. (2 marks)



Taskforce member: one of the people who will decide what the taskforce tells Congress. (2 marks)



Journalist: reporter who covers the taskforce and wants to write a news story about your findings. (2 marks)



Your colleague: analyst at your firm who has run the same analysis on a different sample. (2 marks)



Taskforce analyst: analyst at the taskforce who checks your calculations and runs the same regressions on a separate sample of 500 workers. (2 marks)

Each of these five people will ask you one question during the meeting. Which question you'll get will depend on your results and on your advice to the chair.

After the meeting



The chair

"While you get ready to write your memo, I'm preparing my remarks to Congress. Do you still recommend the finding that you sent me earlier, or do you want to change it? Either way, write me the sentence that I'll use."

The finding that you sent: *the finding that you recommended*

Your reply to the chair (4 marks)

Keep my original finding

Replace it with: *one of your other estimates*

The sentence that you now recommend the chair use, in your own words

0 / 600 characters

What did the meeting confirm or change about your interpretation?

0 / 600 characters

- ☐ I understand that once I send my revision to the chair, my sentence and my reason are locked, even if I go back to an earlier page. (1 mark)

Send the revision to the chair →

YOUR WORK

Hand-off to the memo

The list and the table below set out your results from question 1. In question 2, you will write a memo based on these results. You can return to this page while writing, for reference.

A list of anything missing, with a link back to its page. Once nothing is missing, a list headed “For your memo” gives the estimate in your final recommendation, the other two estimates, your colleague's estimate, and your own estimate from the same regression. Then a table of every number, choice, and reason.

QUESTION 2

The memo

Question 2 of 2. Write a memo to the chair of the taskforce, at most 300 words, using your results on the last page of question 1 (Your work). You can go back to that page while you write, until you click Finish attempt. You don't need a heading, greeting, or sign-off: write only the body of the memo.

In your memo:

- Describe the estimate in your final recommendation to the chair, the one you kept or switched to after the meeting; the list headed “For your memo” on the Your work page gives its value
- Say whom it describes, which groups it compares, and why you chose it over the other two estimates, with units and direction, for example “\$4.36 less per hour”
- If you switched to a different estimate after the meeting, say why
- At the meeting, your colleague compared their estimate from a different random sample of 2,000 workers with your own estimate from the same regression, and both are in the list;

give both estimates and explain what this comparison tells the chair about the estimate in your final recommendation

- Describe one limitation specific to the estimate in your final recommendation or to the data behind it, and say how it affects that estimate
- Make only claims that you can back up with the regressions you ran

Your memo can explain your choice more fully than the reasons you typed in question 1.

Please note that iLearn won't enforce the word limit, so you are responsible for checking your word count before you submit. The 300-word limit covers the whole memo, and only the first 300 words count towards your memo marks.

Attach the Colab notebook that you used, as a .ipynb file (File → Download → Download .ipynb). We will not accept PDFs or screenshots. The notebook is worth 5 marks, and if it does not match your submitted results, you will get 0 marks for it.